MENG WANG

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SUMMARY

I am a Ph.D. candidate in Computer Engineering at the University of British Columbia, working on **scalable and efficient quantum computing systems**. My research spans **quantum architecture, quantum compilation, simulation, and fault-tolerant quantum computing**, with a focus on system-level techniques that reduce the cost of executing quantum programs across the computing stack.

My work has appeared in top computer architecture venues including **ISCA**, **MICRO**, and **ASPLOS**. I develop compiler, architecture, and runtime techniques for quantum circuit simulation, distributed and multi-device execution, and fault-tolerant quantum computation, with an emphasis on bridging theoretical quantum algorithms, practical large-scale systems, and emerging quantum hardware.

I have also built cross-disciplinary and multi-institution collaborations, including quantum chemistry simulation work at UBC with the National Research Council Canada (NRC) and participation in the Quantum Science Center (QSC) and Co-design Center for Quantum Advantage (C2QA) during my research at Pacific Northwest National Laboratory (PNNL). These collaborations connect quantum systems research with applications in computer science, chemistry, physics, and national laboratory programs.

EDUCATION

Ph.D in Electrical and Computer Engineering — *September 2020 - Present*
The University of British Columbia — Vancouver, BC, Canada
Advisor: Prof. Prashant J. Nair

BASc in Computer Engineering — *June 2020*
The University of British Columbia — Vancouver, BC, Canada
Graduated with Distinction

CONFERENCE PUBLICATIONS

1. **ISCA '26** — **Meng Wang**, Chenxu Liu, Samuel Stein, Yufei Ding, Poulami Das, Prashant J. Nair, and Ang Li
   Transpiler-Architecture Co-Design to Curb Clifford Costs in Fault-Tolerant Quantum Computing
   The International Symposium on Computer Architecture, 2026
   Distinguished Artifact Award
2. **ISCA '25** — **Meng Wang**, Swamit Tannu, and Prashant J. Nair
   Accelerating Simulation of Quantum Circuits under Noise via Computational Reuse
   The International Symposium on Computer Architecture, 2025
3. **MICRO '24** — **Meng Wang**, Poulami Das, and Prashant J. Nair
   Qoncord: A Multi-Device Job Scheduling Framework for Variational Quantum Algorithms
   IEEE/ACM International Symposium on Microarchitecture, 2024
4. **ASPLOS '24** — **Meng Wang**, Bo Fang, Ang Li, and Prashant J. Nair
   Red-QAOA: Efficient Variational Optimization through Circuit Reduction
   International Conference on Architectural Support for Programming Languages and Operating Systems, 2024

JOURNAL PUBLICATIONS

1. **TQC '26** Aaron Hoyt, **Meng Wang**, Fei Hua, Chunshu Wu, Chenxu Liu, Muqing Zheng, Samuel Stein, Drew Rebar, Yufei Ding, Travis S. Humble, and Ang Li
   QASMTrans: An End-to-End QASM Compilation Framework with Pulse Generation for Near-Term Quantum Devices
   Transactions on Quantum Computing, accepted, 2026
2. **PRR '26** Sean R. Garner, Nathan M. Myers, **Meng Wang**, Samuel Stein, Chenxu Liu, and Ang Li
   Simulation of Thermal-Relaxation Noise for Quantum Error Correction
   Physical Review Research, **8(2)**, 023134, 2026

ARXIV PREPRINTS

1. **Meng Wang**, Chenxu Liu, Sean Garner, Samuel Stein, Yufei Ding, Prashant J. Nair, and Ang Li
   Tableau-Based Framework for Efficient Logical Quantum Compilation
   arXiv preprint arXiv:2509.02721
2. Chenxu Liu, **Meng Wang**, Samuel A. Stein, Yufei Ding, and Ang Li
   Quantum Memory: A Missing Piece in Quantum Computing Units
   arXiv preprint arXiv:2309.14432
3. Avinash Kumar, **Meng Wang**, Chenxu Liu, Ang Li, Prashant J. Nair, and Poulami Das
   Context Switching for Secure Multi-programming of Near-Term Quantum Computers
   arXiv preprint arXiv:2504.07048
4. Nicholas P. Bauman, Ajay Panyala, Chenxu Liu, Muqing Zheng, **Meng Wang**, and Karol Kowalski
   Quantum Information Harvesting with the Parallel Quantum Flow Algorithm
   arXiv preprint arXiv:2606.04186
5. Sean Garner, Chenxu Liu, **Meng Wang**, Samuel Stein, Ang Li
   STABSim: A Parallelized Clifford Simulator with Features Beyond Direct Simulation
   arXiv preprint arXiv:2507.03092
6. Zeguan Wu, Mingze Li, Muqing Zheng, **Meng Wang**, Junyu Liu, Samuel Stein, Ang Li, Yousu Chen, and Chenxu Liu
   Benchmarking and Resource Analysis for Augmented-Lagrangian Quantum Hamiltonian Descent
   arXiv preprint arXiv:2605.12066
7. Adrian Harkness, Shuwen Kan, Chenxu Liu, **Meng Wang**, John M Martyn, Shifan Xu, Diana Chamaki, Ethan Decker, Ying Mao, Luis F Zuluaga, Tamás Terlaky, Ang Li, and Samuel Stein
   FTCircuitBench: A Benchmark Suite for Fault-Tolerant Quantum Compilation and Architecture
   arXiv preprint arXiv:2601.03185
8. Shuwen Kan, Zefan Du, Chenxu Liu, **Meng Wang**, Yufei Ding, Ang Li, Ying Mao, and Samuel Stein
   SPARO: Surface-code Pauli-based Architectural Resource Optimization for Fault-tolerant Quantum Computing
   arXiv preprint arXiv:2504.21854
9. Fei Hua, Yuwei Jin, Ang Li, Chenxu Liu, **Meng Wang**, Yanhao Chen, Chi Zhang, Ari Hayes, Samuel Stein, Minghao Guo, Yipeng Huang, and Eddy Z. Zhang
   A Synergistic Compilation Workflow for Tackling Crosstalk in Quantum Machines
   arXiv preprint arXiv:2207.05751

10. Ang Li, Chenxu Liu, Samuel Stein, In-Saeng Suh, Muqing Zheng, **Meng Wang**, Yue Shi, Bo Fang, Martin Roetteler, and Travis Humble
TANQ-Sim: Tensorcore Accelerated Noisy Quantum System Simulation via QIR on Perlmutter HPC
arXiv preprint arXiv:2404.13184

WORKSHOP PUBLICATIONS

1. **QCS '23** — **Meng Wang**, Fei Hua, Chenxu Liu, Nicholas Bauman, Karol Kowalski, Daniel Claudino, Travis Humble, Prashant J. Nair, Ang Li
Enabling Scalable VQE Simulation on Leading HPC Systems
The International Conference for High Performance Computing, Networking, Storage, and Analysis Workshop: Quantum Computing Softwares, 2023

2. **QCCC '23** — **Meng Wang**, Bo Fang, Ang Li, and Prashant J. Nair
Efficient QAOA Optimization using Directed Restarts and Graph Lookup
The Second International Workshop on Quantum Classical Cooperative Computing, 2023

3. **Q-CORE** — Yue Shi, Chenxu Liu, Samuel Stein, **Meng Wang**, Muqing Zheng, and Ang Li
Design of an entanglement purification protocol selection module
Q-CORE: Quantum-Classical Orchestration for Resilient Engineering, 2025

INVITED TALKS AND GUEST LECTURES

From Circuits to Systems: Rethinking How Quantum Computation is Realized
◇ Seminar, North Carolina State University — *July 2026*
Host: Prof. Huiyang Zhou — Raleigh, NC, USA

Pushing the Boundaries of NISQ Computing
◇ Guest lecture, The University of Texas at Austin — *November 2024*
Host: Prof. Poulami Das — Austin, TX, USA

Rethinking Variational Quantum Optimization through Circuit-Level Efficiency
◇ Invited talk, University of Wisconsin-Madison — *August 2023*
Host: Prof. Swamit Tannu — Madison, WI, USA

PROFESSIONAL SERVICE

Technical Program Committee — *MICRO 2026*
IEEE/ACM International Symposium on Microarchitecture

RESEARCH SOFTWARE INFRASTRUCTURE

- **NWQEC** – Led development of a C++/Python toolkit for fault-tolerant quantum circuit transpilation, including OpenQASM parsing, Clifford+T synthesis, T-count analysis, Pauli-based circuit optimization, and Clifford reduction passes for QEC-oriented compilation research.
- **NWQ-Sim** – Core contributor to a quantum system simulation environment for multi-node, multi-CPU/GPU HPC systems, supporting state-vector and density-matrix simulation with frontends including QASM, Qiskit, QIR/Q#, and XACC.
- **QASMTrans** – Core contributor to a C++ OpenQASM transpiler for mapping quantum circuits to NISQ devices under vendor-specific basis-gate and qubit-topology constraints, with support for IBMQ, Rigetti, IonQ, and Quantinuum-style backends.

- **XACC** – Contributed to XACC, an extensible hybrid quantum-classical programming and compilation framework with language frontends, backend compilation components, and a quantum intermediate representation for QPUs and simulators.

MENTORING

- Mentored **Aaron Hoyt** from the University of Washington on QASMTrans, contributing to an accepted journal paper at **Transactions on Quantum Computing**.
- Mentored **Sean Garner** from the University of Washington on quantum error-correction noise simulation, contributing to a publication in **Physical Review Research**.
- Mentored **Avinash Kumar** from the University of Texas at Austin on Qontext, a project on secure context switching for multi-programmed near-term quantum computers.

TEACHING EXPERIENCE

Teaching Assistant at UBC — Vancouver, BC, Canada

CPEN 312 - Digital Systems and Microcomputers — *January 2024 - April 2024*
CPSC 322 - Introduction to Artificial Intelligence — *July 2022 - August 2022*
APSC 160 - Introduction to Computation in Engineering Design — *January 2022 - April 2022*
CPEN 411 - Computer Architecture — *September 2021 - December 2021*
CPSC 340 - Machine Learning and Data Mining — *January 2021 - April 2021*

WORK EXPERIENCE

Pacific Northwest National Laboratory — *September 2024 - September 2025*
Ph.D. Research Intern. Mentor: Dr. Ang Li — Richland, WA, USA

- Led research on architectural optimizations for Fault-Tolerant Quantum Computing (FTQC), resulting in an accepted paper at **ISCA'26** focused on transpiler-architecture co-design to curb Clifford costs, with follow-up work currently in preparation for **HPCA'27**.

Pacific Northwest National Laboratory — *October 2022 - August 2023*
Ph.D. Research Intern. Mentor: Dr. Ang Li, Dr. Bo Fang — Richland, WA, USA

- Participated in developing simulation frameworks leveraging leading HPC systems including Frontier, Summit, and Perlmutter to simulate large-scale quantum circuits on GPU clusters. The simulation framework is open-sourced at https://github.com/pnnl/NWQ-Sim.
- Led research improving the efficiency of classical optimization in Quantum Approximated Optimization Algorithm (QAOA), resulting in the publication of Red-QAOA (**ASPLOS '24**), which introduces a novel circuit reduction technique for efficient variational optimization.

AWARDS

Student Travel Grant — *ASPLOS 2024, MICRO 2024, ISCA 2025, ISCA 2026*

Faculty of Applied Science Graduate Award — *2021, 2022, 2023, 2024, UBC*
The annual award, subject to annual review and criteria fulfillment.

APSC Capstone Faculty Award — *May 2020, UBC*

Dean's Honour List — *Jun 2019, UBC*

TECHNICAL SKILLS

Programming languages:
Verilog, C, C++, CUDA, HIP, MPI, Java, Python, ARMv7 Assembly, Racket.
Hardware:
System-on-Chip (SoC), FPGA.